



DECUS

PROGRAM LIBRARY

DECUS NO.	8-280
TITLE	GENERAL SORTING PROGRAM
AUTHOR	M. J. Raymond
COMPANY	Mullards Southampton, Hampshire, England
DATE	March 2, 1970
SOURCE LANGUAGE	PAL III

Although this program has been tested by the contributor, no warranty, express or implied, is made by the contributor, Digital Equipment Computer Users Society or Digital Equipment Corporation as to the accuracy or functioning of the program or related program material, and no responsibility is assumed by these parties in connection therewith.

DECEMBER

1950



GENERAL SORTING PROGRAM

DECUS Program Library Write-up

DECUS No. 8-280

1. Abstract

This programme will sort alpha-numeric records into ascending order. Input-Output is via High Speed Paper Tape peripherals, or the ASR 33 teletype. To improve the use of the available storage space 2 characters are stored per 12 bit computer word. The user must at the time of running, specify the number of character positions to be reserved for each record, and the number of characters to be sorted on. Records longer than the specified length are truncated, records shorter are padded out with blanks.

2. Configuration

- 2.1 PDP-8 series computer with basic 4K memory.
- 2.2 ASR 33 teletype.
- 2.3 High speed paper tape reader (HSR) - optional.
- 2.4 High speed paper tape punch (HP) - optional.

3. Core Requirements

- 3.1 Programme :- 0 to 641
- 3.2 Data :- 642 to 7600.

N.B. The programme sets up a working area one record length from the end of the data area. When the input data attempts to overflow this area, the sort subroutine is automatically entered.

4. Method of Operation

The programme is loaded under the normal procedure with the DEC Binary Loader. When the programme is started, all available storage space is set to zero, the record and sort details are entered via the switch register. The data tape can be read in through the HSR, or teletype reader, the programme will use the HSR if the data tape is waiting in it, if not the programme assumes the teletype is to be used.

Each data record is terminated by a carriage return line feed (cr/lf) and the last record by a cr/lf followed by a dollar sign. The first address where the start of the next record will be stored will always be one record length memory words from the start of the last record. Therefore the ends of records longer than the specification will be overwritten. The sort subroutine is entered when the dollar sign is encountered or when the available storage is filled, record (N) and and record (N + 1) are compared (paragraph 10) if record (N) is lower or equal to record (N + 1) they are left in the same order, record (N + 1) and record (N + 2) will be compared next.

If a move is made backwards N is checked for zero, if it is, it is reset to 1. If a move is made forwards a check is made to see if N is equal to the total number of records, if it is the sort is complete. The output peripheral can be selected from the switch register and can be either the HSP or the teletype. The records are put out in ascending order after a short length of leader. If there are any blank locations at the end of any records they are punched as spaces. This ensures that the output is of fixed record length. Short length of trailer is punched after the end of the last record.

If the continue button is pressed, and a new data tape is ready in the appropriate reader, the sort can be repeated for the same specification.

See also, General Flow Diagram.

5. Operating Procedure

- 5.1 Load the programme with the DEC Binary Loader (See Digital -8-2-U
- 5.2 Load address 200.
- 5.3 Press Start.
- 5.4 Load the data tape into the appropriate tape reader.
- 5.5 Load the record length in octal, onto the switch register.
- 5.6 Press Continue.
- 5.7 Load the number of characters to be sorted on, in octal, onto the switch register.
- 5.8 Press continue.

The data tape will be read in, and the records sorted when the dollar sign is encountered. When the sort is complete, the computer will halt.

- 5.9 Set bit 11 of the switch register to a '1' for HSP output, or to an '0' for ASR 33.
- 5.10 Press Continue.

If the computer halts immediately the sort length entered is greater than the record length, go back to step 5.2

When all records have been put out, the computer will halt. To repeat for new data with the same conditions, press continue. If new conditions are required, goto step 5.2

6. Significant Characters

- ← Erase current record.
- § Commence Sort.

Carriage Return - Terminate current record. (If no data has gone between this cr and the last, this one will be ignored, i.e. no blank record.)

7. Limitations

The following characters are ignored :-

Rub out
Line Feed
Blank Tape.

The @ character is converted to spaces at output.

If the record length or the sort length are odd numbers the higher even number is taken.

8. Execution Time

Large numbers of records do take a significant time to sort, especially if a PDP-8/S is used where the time could possibly extend into minutes.

9. Storage Layout Examples

Example records :-

Lions	103
Elephants	43
Tigers	29

With record length 17 (octal).

Locations

1	2	3	4	5	6	7	10	11	12	13	14	15	16	17	20	21	22	23
LI	ON	S			1	03		EL	EP	HA	NT	S		43		TI	GE	RS

Record 1 points to location 1. Record 2 points to location 11. Record 3 points to location 21.

With Record Length 25 (Octal)

Locations

1	2	3	4	5	6	7	10	11	12	13	14	15	16	17	20	21	22	23
LI	ON	S			1	03					EL	EP	HA	NT	S		43	

Record 1 points to location 1. Record 2 points to location 14.

With Record Length 12 (Octal)

Locations

1	2	3	4	5	6	7	10	11	12	13	14	15	16	17	20	21	22	23
LI	ON	S			1	EL	EP	HA	NT	S		TI	GE	RS				

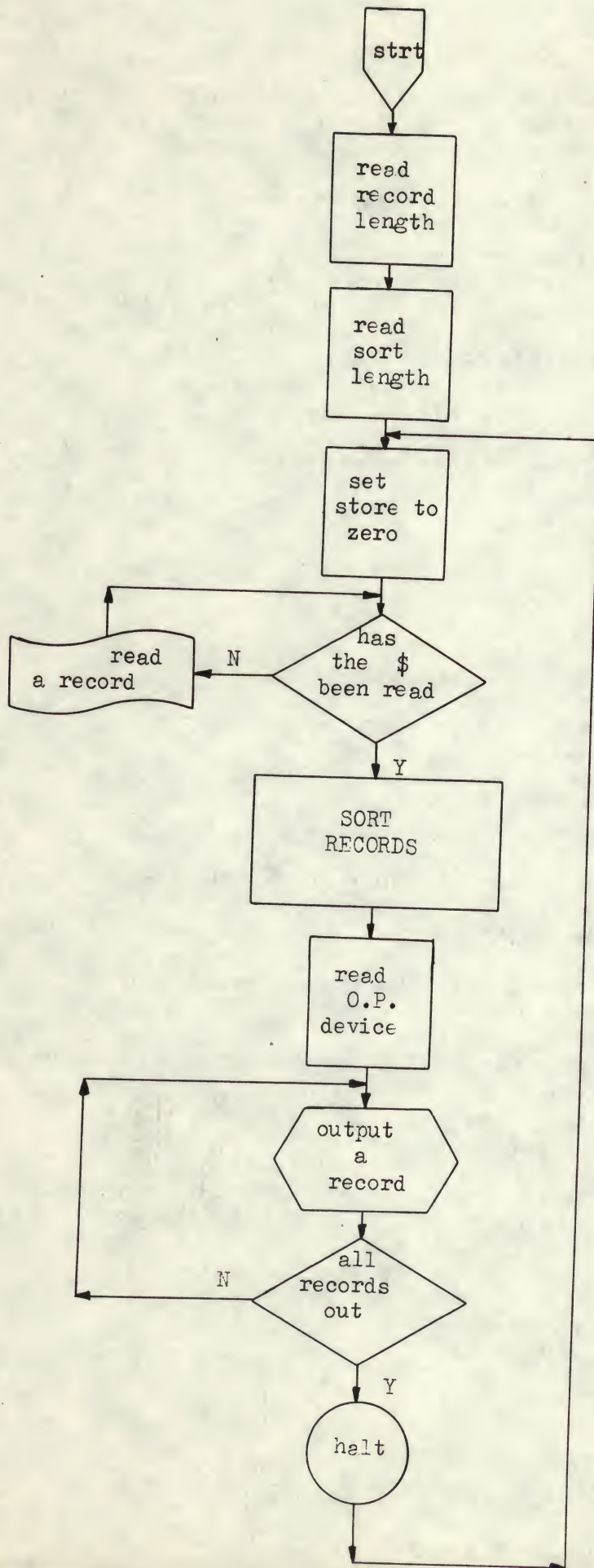
Record 1 points to location 1. Record 2 points to location 7. Record 3 points to location 14.

10. Sort Subroutine - Record Comparison

The sort subroutine compares records by subtracting a word in record (N) from the corresponding word in record (N + 1). If they are equal, it moves to the next word in each record until an unequal answer is reached, or the sort length is expired.

i.e.	1	2	3	4	5	6	
record (N)	TI	GE	RS	2	56		
Octal Equivalent	2411	0705	2223	4062	6566	0000	
Record (N + 1)	TI	GE	RS		43		
Octal Equivalent	2411	0705	2223	4040	6463	0000	

If the sort length was 6, the first 3 words of each record would be compared (all identically equal). Therefore, the records would be left in this order. If the sort length was 12 (10 in decimal) the first 5 words would be compared and at word 4, it would be found that record (N) has a higher value than record (N + 1). Therefore, the records would change order.



General
Flow
Diagram

/ GENERAL SORT PROGRAM FOR PDP-8 SERIES COMPUTERS.
/ M. J. RAYMOND. MULLARDS, SOUTHAMPTON.
/ FEBRUARY 1970.

/
/ *****
/
/ START ADDRESS :- 200.
/ IF HSR IN USE, LOAD DATA TAPE.
/ ENTER RECORD LENGTH ONTO S.R., PRESS CONTINUE.
/ ENTER SORT LENGTH ONTO S.R., PRESS CONTINUE.
/ INPUT DATA. (IF ASR33 IS IN USE.)
/ SET S.R. TO 0001 FOR HSP OUTPUT,
/ 0000 FOR ASR33.
/
/ *****
/

/ PAGE 0.

*1

0001	0000	STORE,	0
0002	0000		0
0003	0000	COUNT,	0
0004	0000	DDI,	0
0005	0000	DELAY,	0
0006	0640	K577,	DATA-1
0007	0641	K600,	DATA
0010	0000	AJID,	0

*20

/ SUBROUTINE TO PRINT HALF A PACKED WORD.

0020	0000	UNIRIM,	0	
0021	0132		AND K77	/ EXTRACT JUNIOR HALF.
0022	7450		SNA	
0023	5037		JMP BLANK	/ DATA = BLANK.
0024	3133		DCA INCOME	
0025	1133		TAD INCOME	
0026	0134		AND K40	/ TEST FOR ALPHA OR NUM.
0027	7650		SNA CLA	
0030	1135		TAD K100	/ ADD BIAS.
0031	1136		TAD K200	
0032	1133		TAD INCOME	
0033	4042	CALL,	JMS OUT	/ PUNCH CHARACTER.
0034	2003		ISZ COUNT	/ ALL CURRENT RECORD OUT ?
0035	5420		JMP I UNIRIM	
0036	5537		JMP I XEOR	/ YES.
0037	7200	BLANK,	CLA	
0040	1154		TAD K240	
0041	5033		JMP CALL	

```

/ SUBROUTINE TO OUTPUT A CHARACTER.
0042 0000 OUT, 0
0043 3133 DCA INCOME
0044 1120 TAD PFLAG
0045 7650 SNA CLA
0046 5055 JMP ITY / TEST FOR OUTPUT DEVICE.
0047 1133 TAD INCOME / HSP.
0050 6026 PLS
0051 6021 PSF
0052 5051 JMP --1
0053 7200 CLA
0054 5442 JMP I OUT / EXIT.
0055 1133 ITY, TAD INCOME
0056 4060 JMS IYPE / TELETYPE.
0057 5442 JMP I OUT / EXIT.

/ SUBROUTINE TO TYPE A CHARACTER.
0060 0000 IYPE, 0
0061 6046 ILS
0062 6041 ISF
0063 5062 JMP --1
0064 7200 CLA
0065 5460 JMP I IYPE

/ SUBROUTINE TO READ A CHARACTER FROM TELETYPE.
0066 0000 SSR, 0
0067 6031 KSF
0070 5067 JMP --1
0071 6036 KRB
0072 5466 JMP I SSR

/ SUBROUTINE TO READ A CHARACTER FROM HSR.
0073 0000 HSR, 0
0074 6014 RFC
0075 6011 RSF
0076 5075 JMP --1
0077 6012 RRB
0100 5473 JMP I HSR

/ SUBROUTINE TO OUTPUT CR-LF.
0101 0000 OCTRLF, 0
0102 1145 TAD K215
0103 4042 JMS OUT
0104 1146 TAD K212
0105 4042 JMS OUT
0106 5501 JMP I OCTRLF

/ SUBROUTINE TO PRINT CR-LF.
0107 0000 CRLF, 0

```

0110	1145		TAD K215
0111	4060		JMS TYPE
0112	1146		TAD K212
0113	4060		JMS TYPE
0114	5507		JMP I CRLF
0115	6000	M2000,	-2000
0116	0600	READX,	READ
0117	0000	RFLAG,	0
0120	0000	PFLAG,	0
0121	0000	RECORD,	0
0122	0000	CONTRL,	0
0123	7741	M37,	-37
0124	1040	K1040,	1040
0125	0200	MIDP,	-7600
0126	0400	SORTII,	SORT
0127	0001	K1,	1
0130	7660	M120,	-120
0131	0000	FLAG,	0
0132	0077	K77,	77
0133	0000	INCOME,	0
0134	0040	K40,	40
0135	0100	K100,	100
0136	0200	K200,	200
0137	0357	XEDR,	EDR
0140	7401	M377,	-377
0141	0133	K133,	133
0142	0317	DOLLAR,	STANBY
0143	0027	K27,	27
0144	0003	K3,	3
0145	0215	K215,	215
0146	0212	K212,	212
0147	0000	RECLN,	0
0150	0000	SORTH,	0
0151	0641	SORTL,	DATA
0152	0000	SORLEN,	0
0153	0000	SORST,	0
0154	0240	K240,	240
0155	7104	XKP,	RAL CLL
0156	7041		CIA
0157	5560		JMP I .+1
0160	0345		NEXT-1
0161	1131	KV2,	TAD FLAG
0162	7650		SNA CLA
0163	5567		JMP I .+4
0164	2121		ISZ RECORD
0165	5566		JMP I .+1
0166	0313		IPIERM+1

0167 0314

IPIERM+2

/ PAGE 1.

*200

0200	6032		KCC	
0201	7300		CLA CLL	
0202	7402		HLT	
0203	7604		LAS	
0204	7001		IAC	
0205	7110		RAR CLL	
0206	3147		DCA RECLN	/ READ RECORD LENGTH FROM SWITCH
0207	7402		HLT	
0210	7604		LAS	
0211	7001		IAC	
0212	7110		RAR CLL	
0213	3152		DCA SORLEN	/ READ SORI KEY LENGTH FROM SR
0214	1147		TAD RECLN	
0215	1125		TAD MTOP	
0216	3125		DCA MTOP	
0217	1006	START,	TAD K577	
0220	3010		DCA AUTO	
0221	1124		TAD K1040	
0222	3003		DCA COUNT	
0223	3410		DCA I AUTO	/ CLEAR ALL DATA AREA.
0224	2003		ISZ COUNT	
0225	5223		JMP --2	
0226	1007		TAD K600	/ RESET STORE.
0227	3001		DCA STORE	
0230	4107		JMS CKLF	
0231	1115		TAD M2000	/ SET TIMER TO TEST HSR.
0232	3005		DCA DELAY	
0233	6014		RFC	
0234	6011	CHECK,	RSF	/ HIGH SPEED READER READY ?
0235	7410		SKP	/ YES !
0236	5242		JMP FAST	
0237	2005		ISZ DELAY	
0240	5234		JMP CHECK	
0241	7410		SKP	
0242	7001	FAST,	IAC	/ NO, IP DEVICE = TELETYPE.
0243	3117		DCA RFLAG	/ 1=HSR, 0=TTY.
0244	3121		DCA RECORD	/ CLEAR RECORD COUNTER.
0245	7040	RSTRT,	CMA	
0246	3122		DCA CONTRL	/ SET WORD PT. TO -1.
0247	3131		DCA FLAG	
0250	1001		TAD STORE	/ COPY NEW RECORD ADDRESS.
0251	3002		DCA STORE+1	
0252	4516	INPUT,	JMS I READX	/ READ A CHARACTER.

0253	1123	TAD M37	/ CHARACTER = "-" ?
0254	7450	SNA	/ YES, LOOSE CURRENT RECORD.
0255	5306	JMP ERASE	
0256	1134	TAD K40	/ CHARACTER = CR.
0257	7650	SNA CLA	/ YES, END OF RECORD.
0260	5312	JMP IPTERM	
0261	1122	TAD CONTRL	/ IP TO SENIOR OR JUNIOR END ?
0262	7650	SNA CLA	
0263	5274	JMP JUNIOR	/ CLEAR IP POINTER.
0264	3122	DCA CONTRL	
0265	1133	TAD INCOME	
0266	7106	CLL RTL	
0267	7006	RTL	
0270	7006	RTL	/ STORE IN SENIOR END.
0271	3401	DCA I STORE	
0272	2131	ISZ FLAG	
0273	5252	JMP INPT	/ ADD NEW JUNIOR TO OLD SENIOR
0274	1133	TAD INCOME	
0275	1401	TAD I STORE	/ INCREMENT STORE.
0276	3401	DCA I STORE	
0277	2001	ISZ STORE	
0300	7100	CLL	
0301	1001	TAD STORE	/ STORE FULL ?
0302	1125	TAD MIDP	/ NO.
0303	7620	SNL CLA	/ YES, ENTER SORT.
0304	5374	JMP PRESET	
0305	5317	JMP STANBY	
0306	7200	CLL	
0307	1002	TAD STORE+1	
0310	3001	DCA STORE	
0311	5245	JMP RSTRT	
0312	5161	JMP KVR	/ MOVE STORE FOR NEXT RECORD.
0313	1147	TAD RECLN	
0314	1002	TAD STORE+1	
0315	3001	DCA STORE	
0316	5245	JMP RSRT	
0317	1002	TAD STORE+1	/ COPY END OF STORE FOR SORT
0320	3150	DCA SORTH	/ ENTER SORT SUBROUTINE.
0321	4526	JMS I SORTIT	
0322	7402	HLT	/ READ REQUIRED OP DEVICE FROM
0323	7604	LAS	
0324	0127	AND K1	/ 1=HSP, 0=ITY.
0325	3120	DCA PFLAG	
0326	1130	TAD M120	
0327	3003	DCA COUNT	/ PUNCH LEADER.
0330	4042	JMS OUT	
0331	2003	ISZ COUNT	
0332	5330	JMP --2	/ PUNCH CR-LF.
0333	4101	JMS OCRLF	
0334	1121	TAD RECORD	

0335	7041		CIA	
0336	3121		DCA RECORD	
0337	1007		TAD K600	/ COMPLIMENT NO. OF RECORDS.
0340	3001		DCA STORE	
0341	1001	INDEX,	TAD STORE	
0342	3002		DCA STORE+1	/ COPY STORE.
0343	1147		TAD RECLN	/ SET UP WORD COUNT.
0344	5155		JMP XKP	
0345	3003		DCA COUNT	
0346	1401	NEXT,	TAD I STORE	
0347	7112		CLL RTR	
0350	7012		RTR	
0351	7012		RTR	/ ROTATE SENIOR FOR O.P.
0352	4020		JMS UNTRIM	
0353	1401		TAD I STORE	/ PUNCH.
0354	4020		JMS UNTRIM	/ SET JUNIOR.
0355	2001		ISZ STORE	/ PUNCH.
0356	5346		JMP NEXT	
0357	4101	FOR,	JMS OCKLF	
0360	1147		TAD RECLN	/ PUNCH CR-LF.
0361	1002		TAD STORE+1	/ SET STORE TO NEW RECORD.
0362	3001		DCA STORE	
0363	2121		ISZ RECORD	/ INCREMENT RECORD COUNT.
0364	5341		JMP INDEX	
0365	1130		TAD M120	/ ALL RECORDS OUT !
0366	3003		DCA COUNT	
0367	4042		JMS OUT	/ PUNCH TRAILER.
0370	2003		ISZ COUNT	
0371	5367		JMP --2	
0372	7402		HLT	/ END.
0373	5217		JMP START	/ REPEAT FOR SAME CONDITIONS.
0374	7040	PRESET,	CMA	
0375	3122		DCA CONTRL	
0376	5252		JMP INPUT	

/ SORT SUBROUTINE FOR FIXED NUMBER OF RECORDS.
 / RECLN = RECORD LENGTH.
 / SORLEN = NUMBER OF WORDS PER RECORD TO BE SORTED ON.
 / SORTL = LOC. CONTAINING 1 ST. WORD OF 1ST. RECORD.
 / SORTH = LOC+1 CONTAINING LAST WORD OF LAST RECORD.

/ PAGE 2.

*400

0400	0000	SORT,	0
0401	7200		CLA
0402	1152		TAD SORLEN
0403	7041		CIA
0404	1147		TAD RECLN

0405 7710
 0406 7402
 0407 1151 SET,
 0410 3365
 0411 1151
 0412 1147
 0413 7000
 0414 3367
 0415 1150
 0416 3363
 0417 1147
 0420 7041
 0421 3364
 0422 5230
 0423 1151
 0424 7041
 0425 1365
 0426 7710
 0427 5207
 0430 1365 GEREDY,
 0431 3366
 0432 1365
 0433 1153
 0434 3365
 0435 1367
 0436 3370
 0437 1367
 0440 1153
 0441 3367
 0442 1152
 0443 7041
 0444 3372
 0445 1765 COMPR,
 0446 7141
 0447 1767
 0450 7450
 0451 5335
 0452 7630
 0453 5341
 0454 1364
 0455 3371
 0456 4355 CHANGE,
 0457 1765
 0460 3763
 0461 2365
 0462 2363
 0463 2371
 0464 5257
 0465 1364
 0466 1365

SPA CLA
 HLT
 TAD SORTL
 DCA REC1
 TAD SORTL
 TAD RECLEN
 NOP
 DCA REC2
 TAD SORTH
 DCA BUFFA
 TAD RECLEN
 CIA
 DCA MRELEN
 JMP GEREDY
 TAD SORTL
 CIA
 TAD REC1
 SPA CLA
 JMP SET
 TAD REC1
 DCA REC1+1
 TAD REC1
 TAD SORTST
 DCA REC1
 TAD REC2
 DCA REC2+1
 TAD REC2
 TAD SORTST
 DCA REC2
 TAD SORLEN
 CIA
 DCA ZERCOIT
 TAD I REC1
 CIA CLL
 TAD I REC2
 SNA
 JMP ZERO
 SZL CLA
 JMP OK
 TAD MRELEN
 DCA COUNTA
 JMS RESET
 TAD I REC1
 DCA I BUFFA
 ISZ REC1
 ISZ BUFFA
 ISZ COUNTA
 JMP CHANGE+1
 TAD MRELEN
 TAD REC1

/ SORLEN > RECLEN
 / SET UP CURRENT RECORD POINTER
 / SET UP CURRENT RECORD+1 POINTER
 / SET UP WORK AREA.
 / SET UP RECORD ELEMENT COUNT
 / COPY CURRENT RECORD ADDRESS
 / COMPARE RECORD AND RECORD+1.
 / ELEMENTS WITHIN RECORDS ARE
 / RECORDS ARE IN THE RIGHT ORDER
 / RECORDS ARE IN THE WRONG ORDER
 / CHANGE ROUND RECORD AND RECORD

0467	3365	
0470	1364	
0471	1363	
0472	3363	
0473	1364	
0474	3371	
0475	1767	MOOV,
0476	3765	
0477	2367	
0500	2365	
0501	2371	
0502	5275	
0503	1364	
0504	1365	
0505	3365	
0506	1364	
0507	1367	
0510	3367	
0511	1364	
0512	3371	
0513	1763	COPIE,
0514	3767	
0515	2367	
0516	2363	
0517	2371	
0520	5313	
0521	1364	
0522	1367	
0523	3367	
0524	1364	
0525	1363	
0526	3363	
0527	1365	
0530	3367	
0531	1365	
0532	1364	
0533	3365	
0534	5223	
0535	2365	ZERO,
0536	2367	
0537	2372	
0540	5245	
0541	4355	OK,
0542	1367	
0543	3365	
0544	1367	
0545	1147	
0546	3367	
0547	1150	
0550	7141	

DCA REC1
 TAD MRELEN
 TAD BUFFA
 DCA BUFFA
 TAD MRELEN
 DCA COUNTA
 TAD I REC2
 DCA I REC1
 ISZ REC2
 ISZ REC1
 ISZ COUNTA
 JMP MOOV
 TAD MRELEN
 TAD REC1
 DCA REC1
 TAD MRELEN
 TAD REC2
 DCA REC2
 TAD MRELEN
 DCA COUNTA
 TAD I BUFFA
 DCA I REC2
 ISZ REC2
 ISZ BUFFA
 ISZ COUNTA
 JMP COPIE
 TAD MRELEN
 TAD REC2
 DCA REC2
 TAD MRELEN
 TAD BUFFA
 DCA BUFFA
 TAD REC1
 DCA REC2
 TAD REC1
 TAD MRELEN
 DCA REC1
 JMP GEREDY-5
 ISZ REC1
 ISZ REC2
 ISZ ZERCOI
 JMP COMPR
 JMS RESET
 TAD REC2
 DCA REC1
 TAD REC2
 TAD RECLEN
 DCA REC2
 TAD SORTH
 CIA CLL

/ MOVE POINTER TO NEXT RECORD
 / (WITHIN SORT KEY)
 / RECORDS ARE IN THE RIGHT ORR
 / MOVE TO NEXT PAIR.

0551	1367		IAD REC2
0552	7630		SZL CLA
0553	5600		JMP I SORT
0554	5230		JMP GEREDY

/ RESET RECORD POINTERS.

0555	0000	RESET,	0
0556	1366		TAD REC1+1
0557	3365		DCA REC1
0560	1370		TAD REC2+1
0561	3367		DCA REC2
0562	5755		JMP I RESET

0563	0000	BUFFA,	0
0564	0000	MRELEN,	0
0565	0000	REC1,	0
0566	0000		0
0567	0000	REC2,	0
0570	0000		0
0571	0000	COUNTA,	0
0572	0000	ZERCOT,	0

/ PAGE 3.
*600

/ SUBROUTINE TO READ A CHARACTER.

0600	0000	READ,	0
0601	7200		CLA
0602	1117		TAD RFLAG
0603	7650		SNA CLA
0604	5234		JMP SLOW
0605	4073		JMS HSR
0606	3133	OKIN,	DCA INCOME
0607	1133		TAD INCOME
0610	7450		SNA
0611	5201		JMP READ+1
0612	1140		TAD M377
0613	7450		SNA
0614	5201		JMP READ+1
0615	1141		TAD K133
0616	7450		SNA
0617	5542		JMP I DOLLAR
0620	1143		TAD K27
0621	7450		SNA
0622	5236		JMP LOOSE
0623	1144		TAD K3
0624	7650		SNA CLA
0625	5201		JMP READ+1

/ CHARACTER = 0 ?

/ CHARACTER = RUBBOUT ?

/ CHARACTER = \$?

/ CHARACTER = CR ?

/ CHARACTER = LF ?

0626	7200	HOLD,	CLA
0627	1133		IAD INCOME
0630	0132		AND K77
0631	3133	HOP,	DCA INCOME
0632	1133		IAD INCOME
0633	5600		JMP I READ
0634	4066	SLOW,	JMS SSR
0635	5206		JMP OKIN
0636	7040	LOOSE,	GMA
0637	5231		JMP HOP
0640	0000		0
0641	0000	DATA,	0

AUTO	0010
BLANK	0037
BUFFA	0563
CALL	0033
CHANGE	0456
CHECK	0234
COMPR	0445
CONTRL	0122
COPIE	0513
COUNT	0003
COUNTA	0571
CRLF	0107
DATA	0641
DELAY	0005
DOLLAR	0142
EOR	0357
ERASE	0306
FAST	0242
FLAG	0131
GEREDY	0430
HOLD	0626
HOP	0631
HSR	0073
INCOME	0133
INDEX	0341
INPUT	0252
IPTERM	0312
JUNIOR	0274
KV2	0161
K1	0127
K100	0135
K1040	0124
K133	0141
K200	0136
K212	0146
K215	0145
K240	0154

K27	0143
K3	0144
K40	0134
K577	0006
K600	0007
K77	0132
LOOSE	0636
MOOV	0475
MRELEN	0564
MTOP	0125
M120	0130
M2000	0115
M37	0123
M377	0140
NEXT	0346
OCRLF	0101
ODT	0004
OK	0541
OKIN	0606
OUT	0042
PFLAG	0120
PRESET	0374
READ	0600
READX	0116
RECLN	0147
RECORD	0121
REC1	0565
REC2	0567
RESET	0555
RFLAG	0117
RSTRT	0245
SET	0407
SLOW	0634
SORLEN	0152
SORST	0153
SORT	0400
SORTH	0150
SORTIT	0126
SORTL	0151
SSR	0066
STANBY	0317
START	0217
STORE	0001
ITY	0055
ITYE	0060
UNTRIM	0020
XEDR	0137
XKP	0155
ZERCO I	0572
ZERO	0535